



Azure at the Edge: Real-World Use Cases

Azure at the Edge

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Azure At The Edge Real World Use Cases

- Current Edge Products
- Azure Stack HCI Hosting AVD Session Hosts
- Azure Stack Hub – Azure in your Datacenter
- Azure Stack Edge - Connectivity
- Azure Local with Disconnected Operations (Preview)

Current Azure Stack Portfolio

Consistently build and run hybrid apps across on-premises, cloud, and edge



Azure Stack Edge

Cloud-managed appliance



Azure Stack HCI

Hyperconverged solution

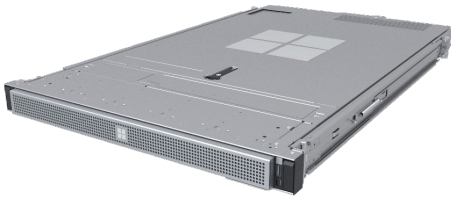


Azure Stack Hub

Cloud-native integrated system

Azure Stack Edge

Commercial



Enterprise-ready server with the option for one or two NVIDIA T4 Tensor Core GPUs

Rugged



Ruggedized, portable, hardware-accelerated form-factors for harsh field conditions



Azure Stack Hub



4 Node Ruggedized



8 Node Dense



16 Node Hybrid

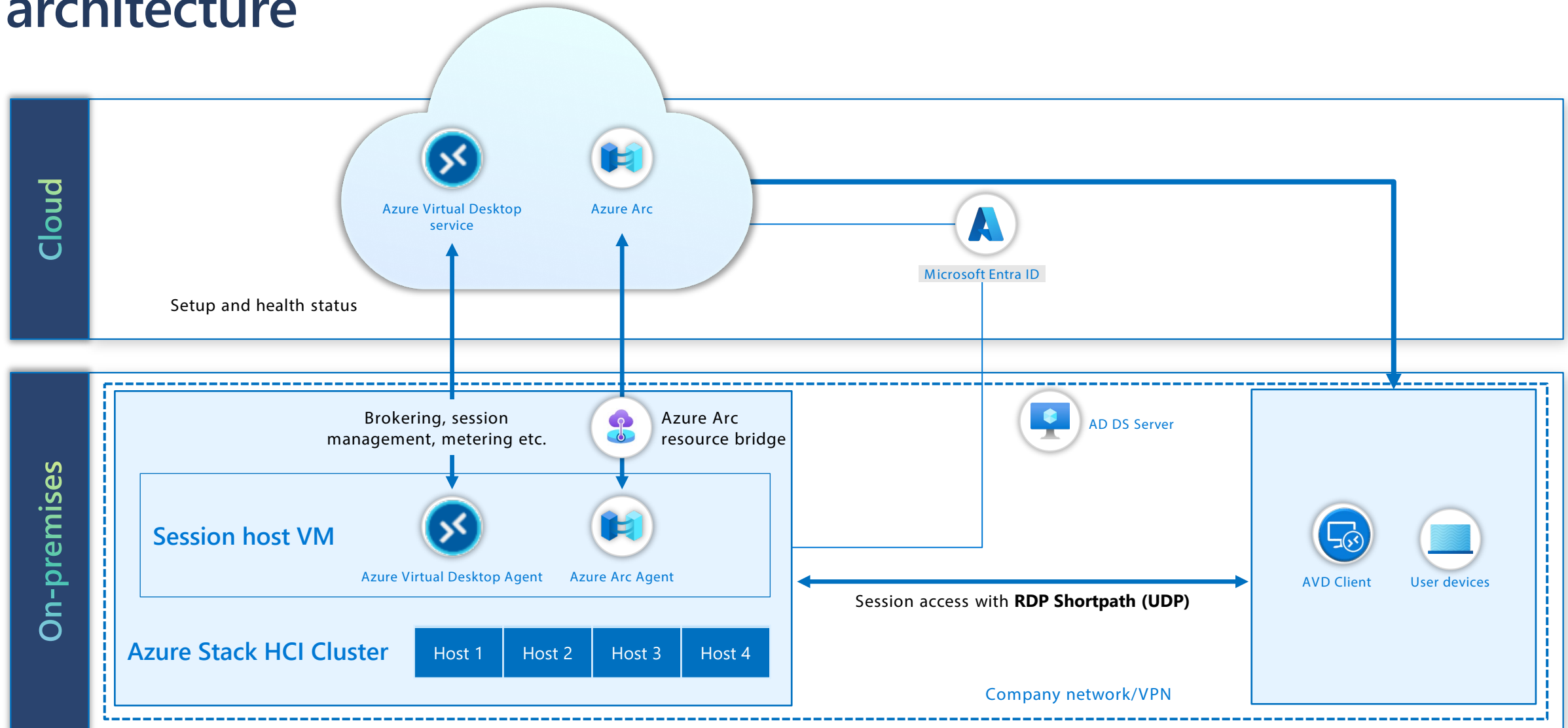
Azure Stack HCI



Customer Requirements

- Improve performance for Azure Virtual Desktop users in areas with poor connectivity to the Azure public cloud by giving them session hosts closer to their location.
- Meet data locality requirements by keeping app and user data on-premises.
- Improve access to legacy on-premises apps and data sources by keeping desktops and apps in the same location.

Azure Virtual Desktop for Azure Stack HCI high-level solution architecture



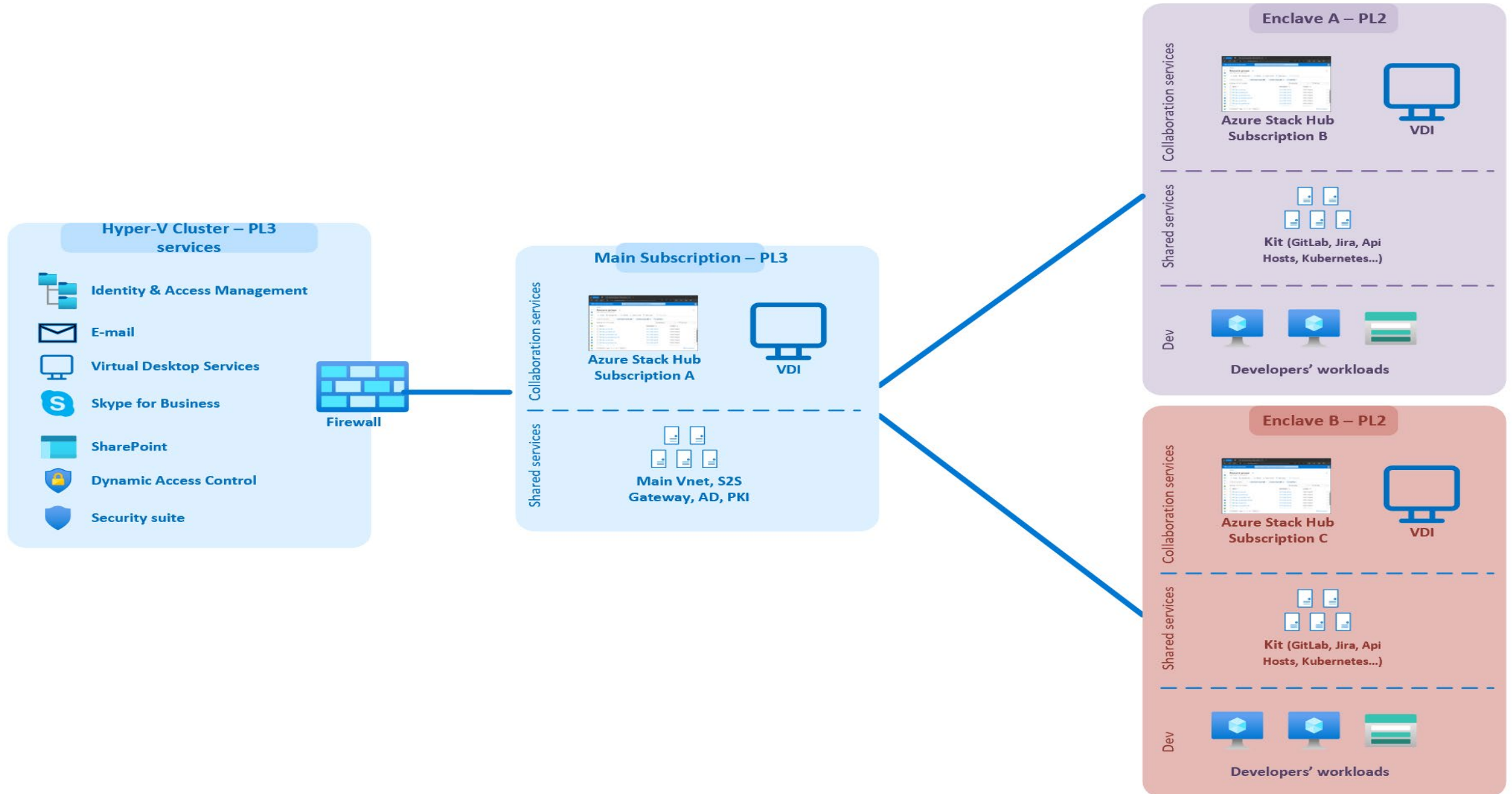
Customer Requirements

- Ability to provision & provide PL3 services to development teams
- Allow development teams to self-provision IaaS Virtual Machines, Containers and Virtual Machine Scale Sets
- Must be able to operator completely disconnected
- Must prevent unauthorized movements of data in between development environments

Understanding Azure Stack Hub

- Azure Stack Hub is an extension of Azure services to on-premises environments.
- It allows businesses to build and deploy applications consistently across cloud and local infrastructure.
- Azure Stack Hub supports a range of services, including IaaS and PaaS.
- It enables disconnected operations, providing flexibility when connectivity is limited.
- Ideal for organizations needing compliance with data residency regulations.

Azure Stack Hub & Hyper-V Cluster

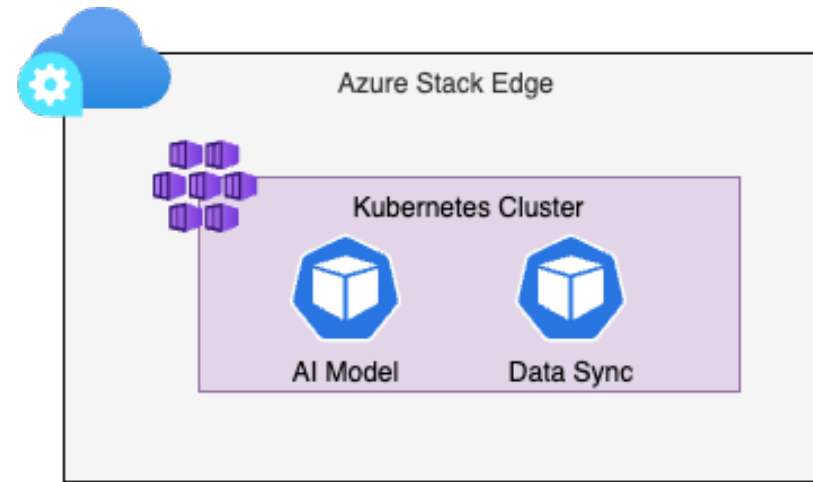
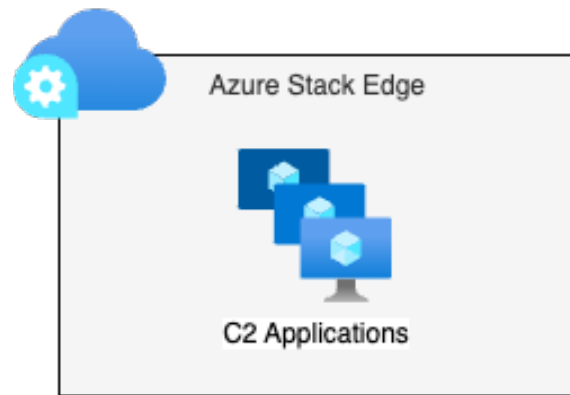


Edge Connectivity Patterns

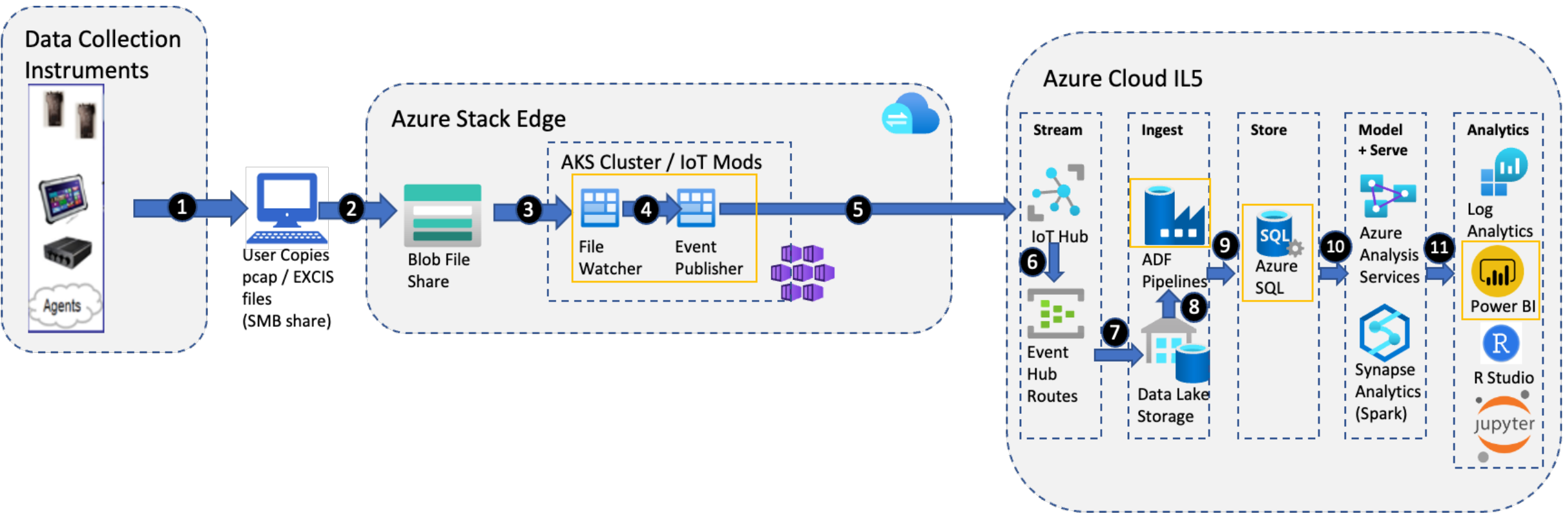
- Edge Devices as standalone services
- Edge Devices running workloads and aggregating in Azure
- Edge Devices as an extension of your cloud architecture



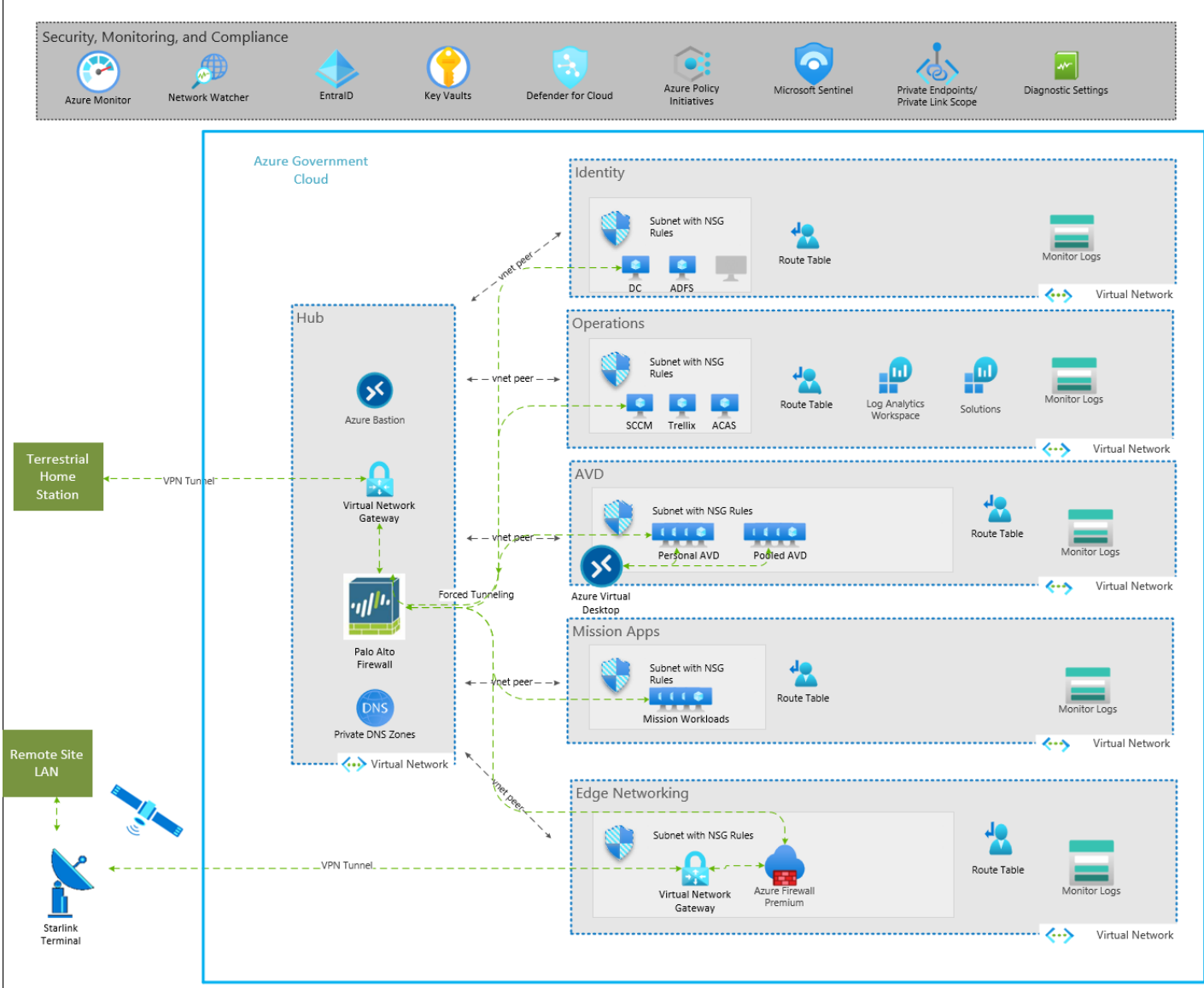
Standalone Edge Capabilities



Edge Workload Aggregation



Extending Cloud to the Edge



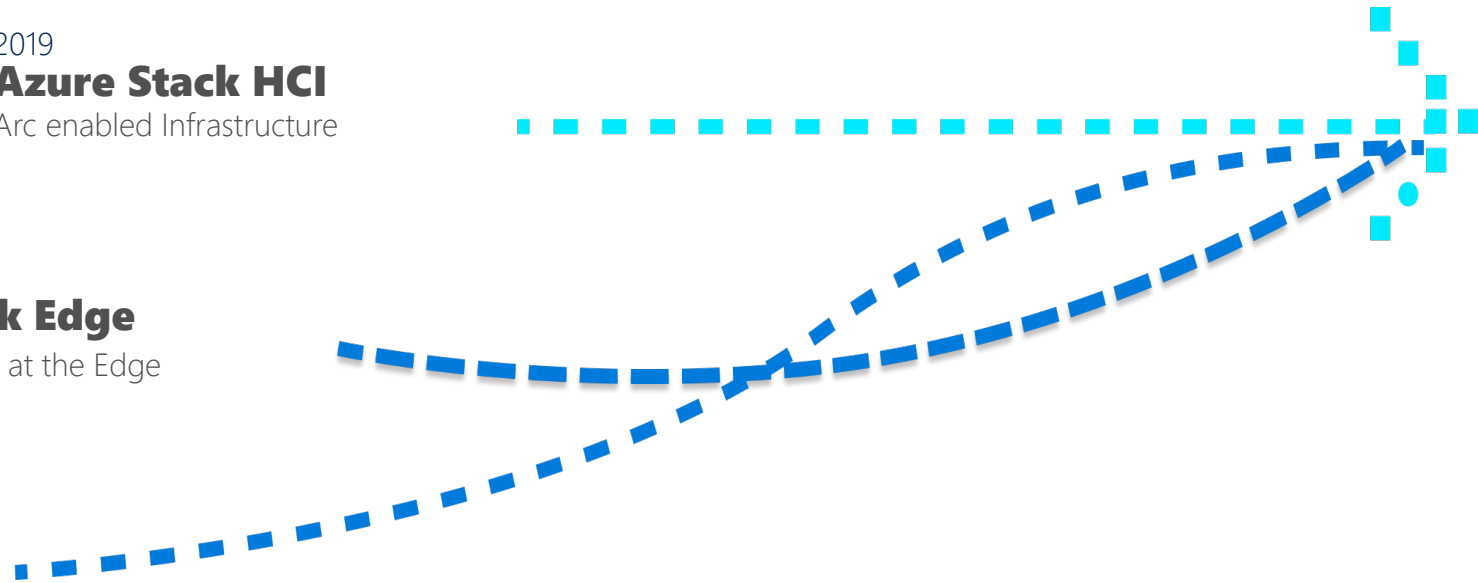
Azure Local

2019
Azure Stack HCI
Arc enabled Infrastructure

2018
Azure Stack Edge
Compute, AI, ML at the Edge

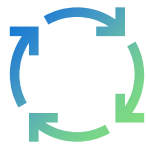
2016
Azure Stack Hub
Autonomous Cloud

2025
Azure Stack HCI
Arc enabled Infrastructure



Introducing Azure Local

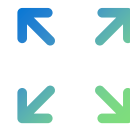
Cloud infrastructure for distributed locations, enabled by Azure Arc



Operate and scale
with the power of
the cloud



Ready for all your
apps: VMs and
containers alike



Flexibility to meet
your requirements
and budget



Extend cloud
security to your
distributed locations

Understanding Azure Local

- Azure Local is designed to provide near-cloud capabilities on-premises.
- It supports various workloads, including VMs and containerized applications.
- Azure Local enables seamless connectivity with Azure cloud services.
- It enhances operational flexibility in environments with limited internet access.
- Azure Local is ideal for edge computing scenarios, improving response times.

Consistent Azure & Arc capabilities, even disconnected

Monitoring

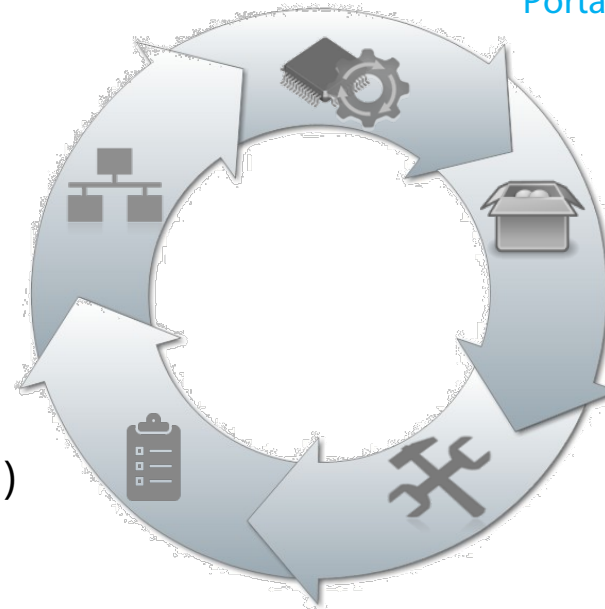
- Monitor standard metrics for the system (Appliance VM, HCI, AKS & VMs)

Management Interfaces

- Manage the system (VM appliance, HCI clusters, AKS & VMs)
- Choice of API (ARM), CLI (PowerShell), or (UI) Azure Portal to manage

Manage Secrets

- Store and retrieve keys, secrets and certificates via Key Vault



Policy Management

- Enforce Azure policies on resource creation

Manage Azure Kubernetes Service (AKS)

- Deploy and configure AKS clusters
- Deploy containers
- Local Container registry

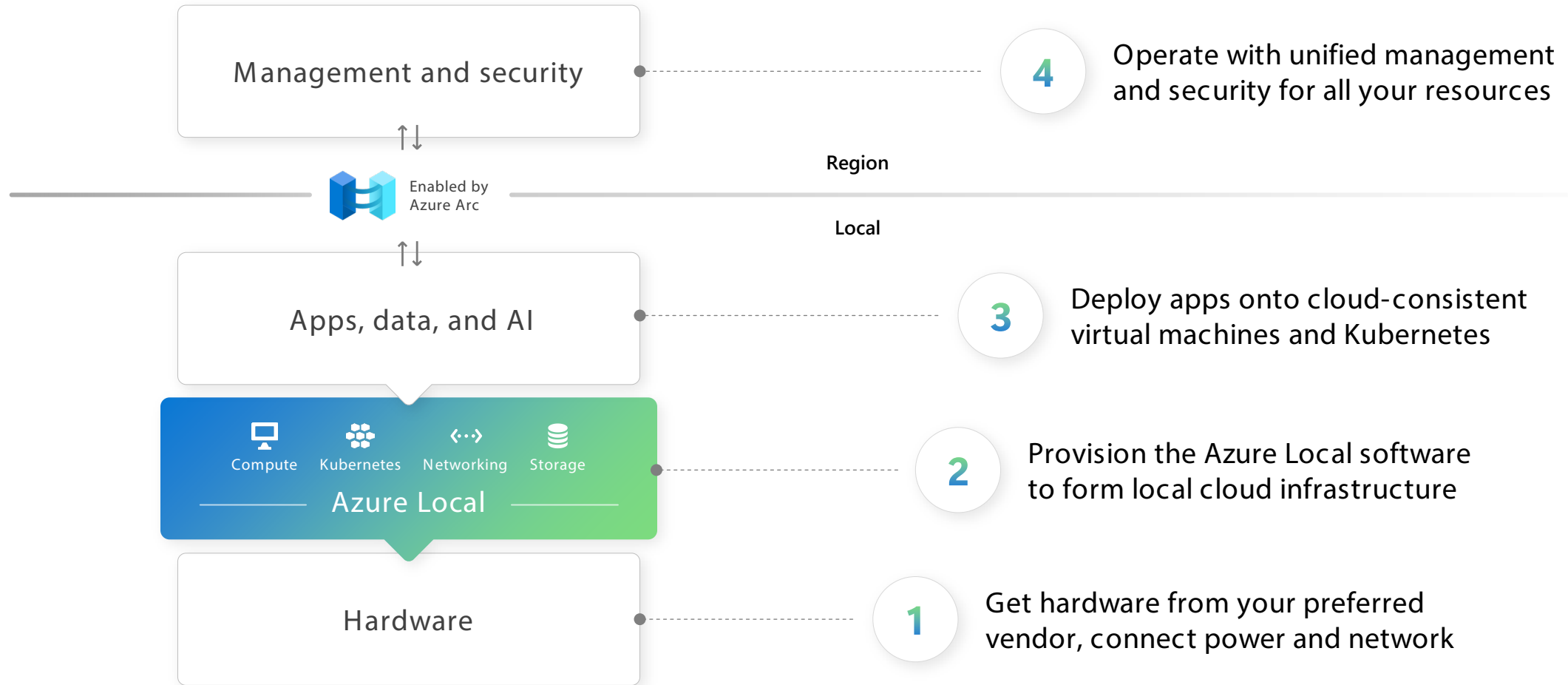
Deploy & Manage HCI Platform

- Deploy & create the cluster
- Apply updates
- Add/remove nodes

Manage Virtual Machines

- Create, modify, and delete VMs

How Azure Local works (connected)



Introducing disconnected operations (preview)



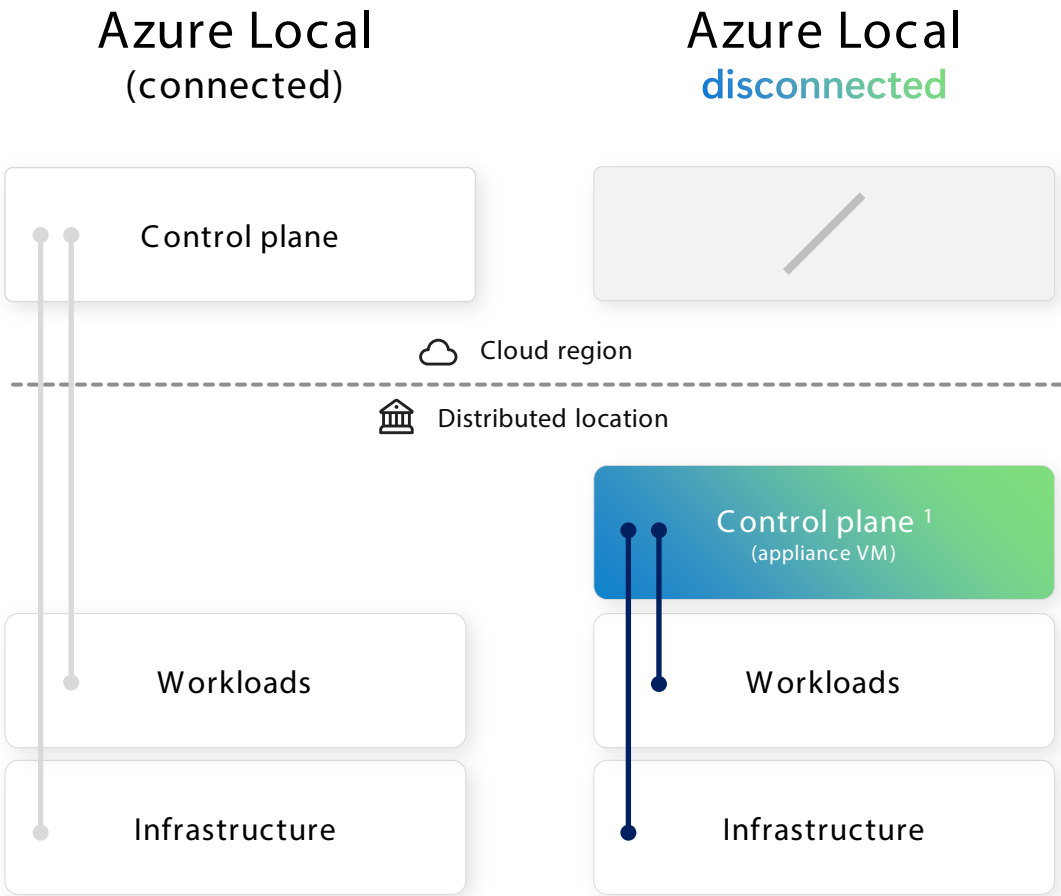
Satisfy regulatory requirements by operating permanently disconnected from the cloud



Host backend Azure resource manager, portal, and services in local appliance VM

Subset of Azure services available³:

 Portal	 ARM	 Copilot	 Key Vaults	 Policy ²	 Defender
 Local	 Machines	 Kubernetes	 Registries	 AVD	 Others



¹ : Available only to customers who prequalify based on industry, use case, and other considerations ² : Partial functionality ³ : Datacenter services required for integration

All resources - Azure Local

https://portal.autonomous.cloud.private/#view/HubsExtension/BrowseAllResourcesBlade/resourceType/Microsoft.Resources%2Fresources

Azure Local

Search resources

Create a resource

Dashboard

All services

FAVORITES

All resources

Resource groups

Subscriptions

Identity management

Resource Explorer

Tags

Marketplace

Container registries

Azure Arc

Machines - Azure Arc

Kubernetes - Azure Arc

Azure Local

Key vaults

Dashboard >

All resources

Azure Local with disconnected operations

Create Edit columns Refresh Export to CSV Delete

Subscriptions: All 3 selected – Don't see a subscription? Open Directory + Subscription settings

Filter by name... All subscriptions All resource groups All types All tags

1 of 20 items selected Show hidden types

Name	Type	Resource group	Location
aksarc-lnet-static	Azure Local Logical network	infrastructure-rg	Autonomous
arcvm-lnet-static	Azure Local Logical network	infrastructure-rg	Autonomous
arcvm-vnic	Azure Local VM Network Interface	infrastructure-rg	Autonomous
edgeartifacts	Container registry	shared-rg	Autonomous
s-cluster	Azure Local	infrastructure-rg	Autonomous
s-cluster-arcbridge	Resource bridge	infrastructure-rg	Autonomous
s-cluster-customlocation	Custom location	infrastructure-rg	Autonomous
sclusterkv1sbrphic	Key vault		Autonomous
testACR	Container registry		Autonomous
test-gallery-image	Azure Local Gallery image	infrastructure-rg	Autonomous

Azure Local with disconnected operations

Watch the demo

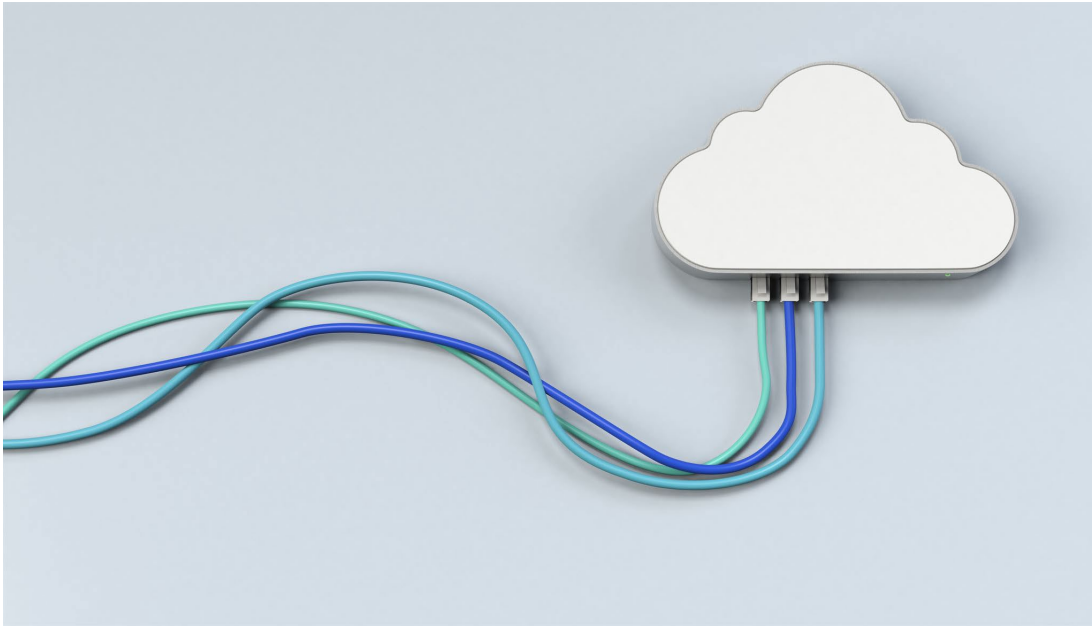
Disconnected operations
in action

Azure Local



<https://aka.ms/disconnected-operations-demo>
(2 mins 23 secs)

Why Switch from VMware to Azure Local?

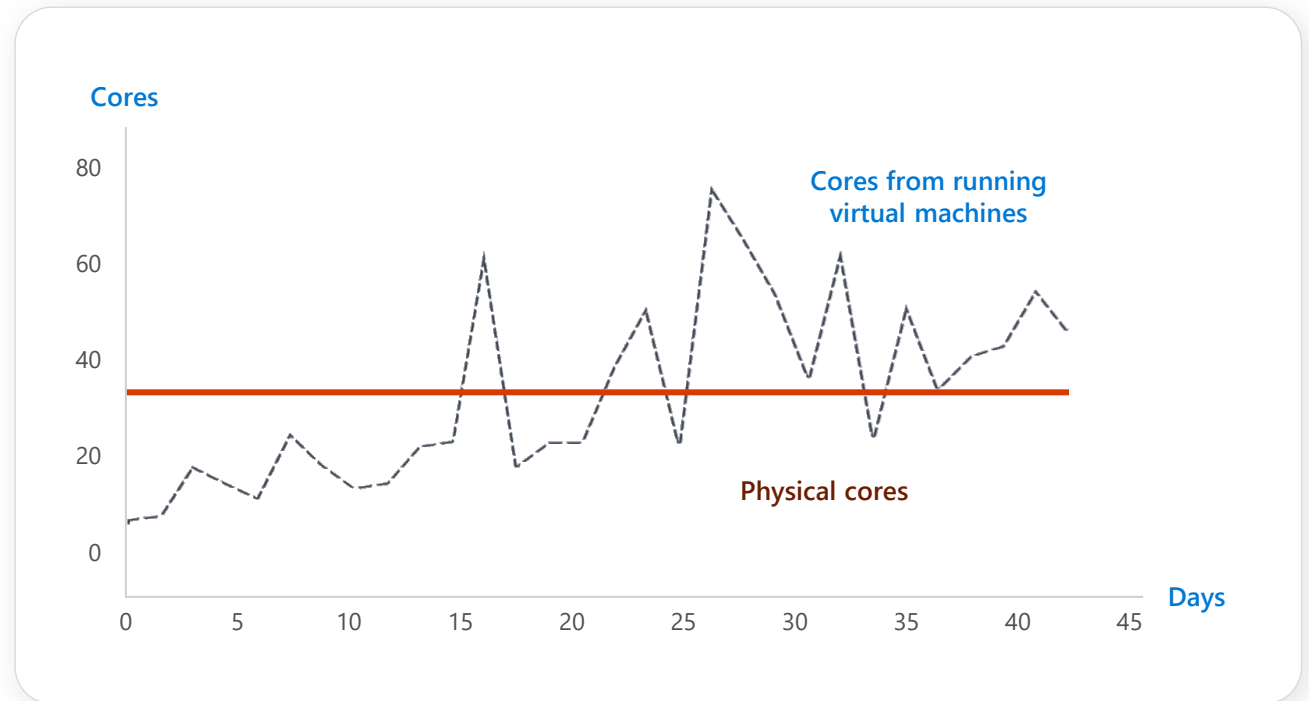


- Azure Local offers better cost efficiency compared to VMware.
- Enhanced flexibility and scalability with Azure's cloud solutions.
- Seamless integration with existing Microsoft services and infrastructure.

What does Azure Local charge for?

\$10 USD/month
per physical processor core*

*Does not include
Guest OS licensing



Predictable

Doesn't vary with consumption,
more VMs doesn't cost more



Simple

No math with memory, storage,
or network ingress/egress



Efficient

Virtualize efficiently, with higher
v-to-p density

Migrate from VMware to Azure Local (preview)



Reduce your VMware costs and footprint with full-stack alternative



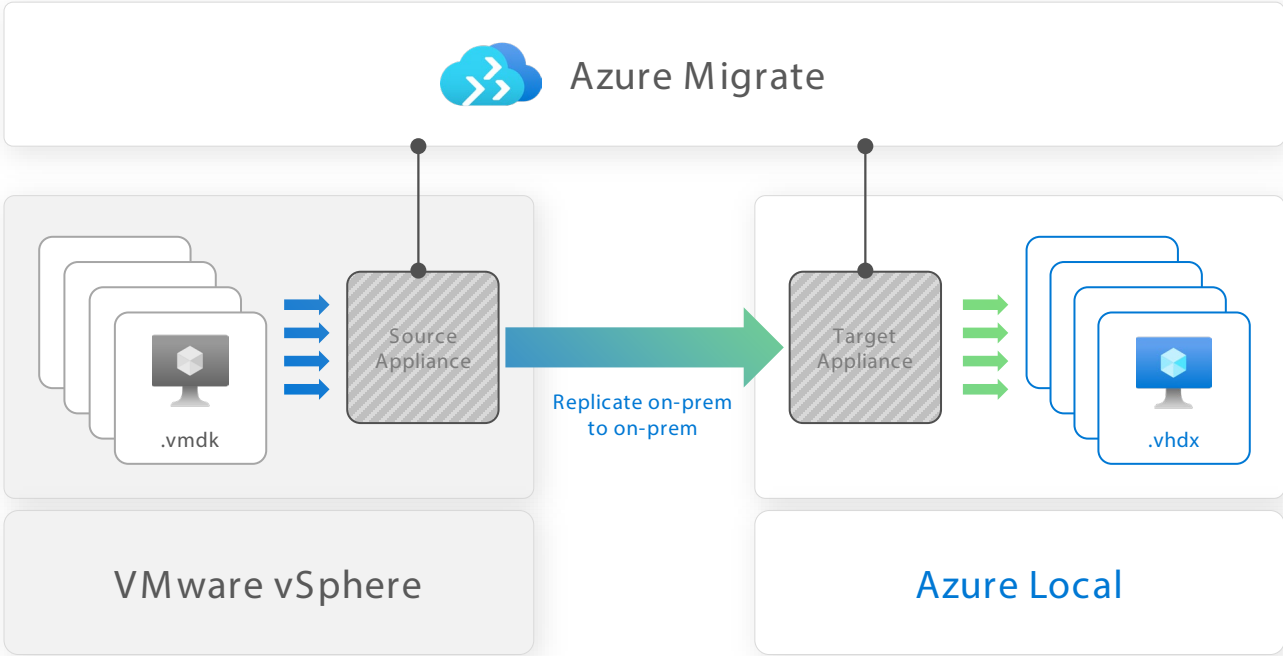
No need to change/rewrite apps



Copy and convert VMware VMDK to Azure Arc VM entirely on-premises



Guided workflow in Azure Migrate



Thank You!

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